

Multiple Regression

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Regression

Regression is the broad idea, and simple linear regression, multiple linear regression, and logistic regression are different types of regression models.

- **Simple linear regression** has one predictor and one outcome.
- **Multiple linear regression** has several predictors and one outcome.
- **Logistic regression** is used when the outcome is categorical, usually binary.

In multiple regression, we use several predictors to explain or predict one outcome.

Example 1: Trees

We will use the built-in `trees` dataset.

Data Visualisation

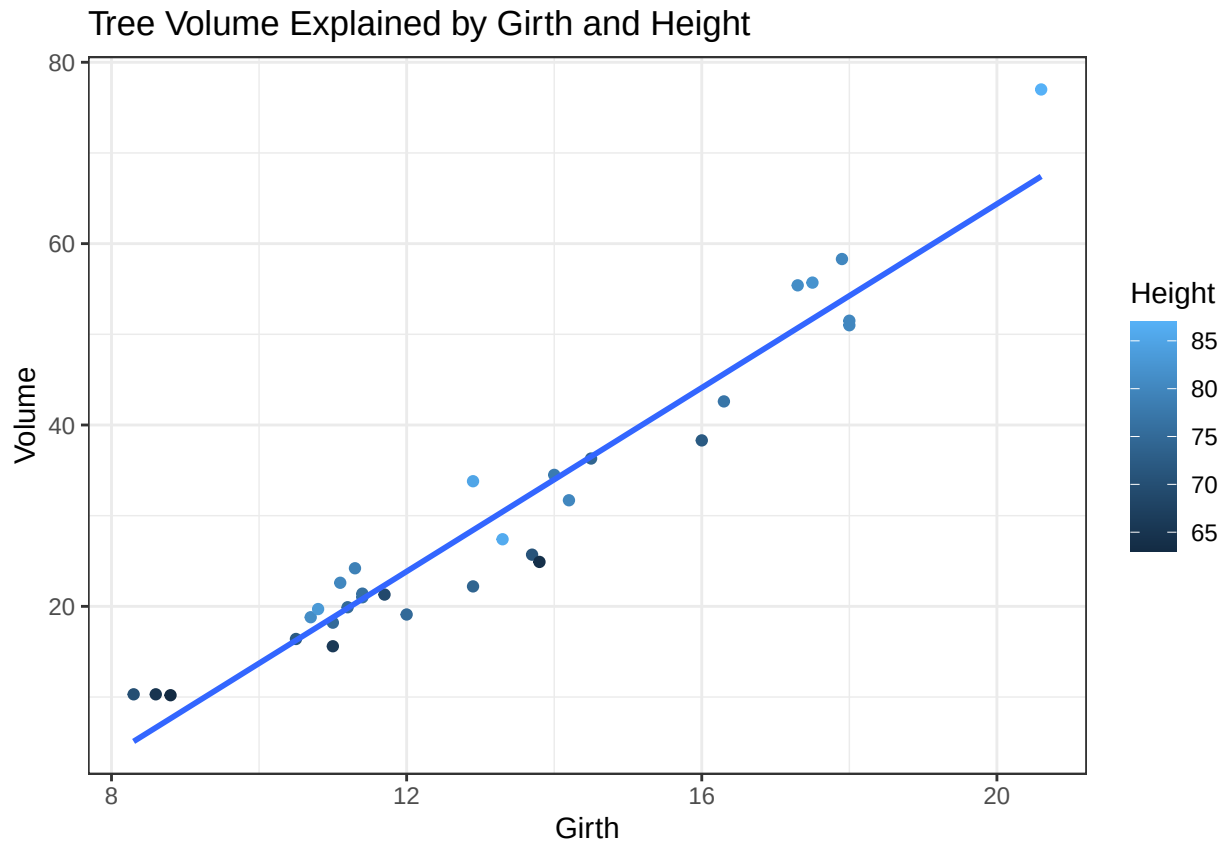
```
library(ggplot2)

data(trees)

trees |>
  ggplot(aes(Girth, Volume, color = Height)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  theme_bw() +
  labs(
    title = "Tree Volume Explained by Girth and Height",
    x = "Girth",
    y = "Volume",
    color = "Height"
  )
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: The following aesthetics were dropped during statistical transformation:
## colour.
## i This can happen when ggplot fails to infer the correct grouping structure in
## the data.
## i Did you forget to specify a 'group' aesthetic or to convert a numerical
## variable into a factor?
```



Model

We can use multiple linear regression to model tree volume using girth and height as predictors.

```
model <- lm(Volume ~ Girth + Height, data = trees)
summary(model)
```

```
##
## Call:
## lm(formula = Volume ~ Girth + Height, data = trees)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -6.4065 -2.6493 -0.2876  2.2003  8.4847
##
## Coefficients:
```

```
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept) -57.9877      8.6382  -6.713 2.75e-07 ***
## Girth        4.7082      0.2643  17.816 < 2e-16 ***
## Height       0.3393      0.1302   2.607  0.0145 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.882 on 28 degrees of freedom
## Multiple R-squared:  0.948, Adjusted R-squared:  0.9442
## F-statistic: 255 on 2 and 28 DF, p-value: < 2.2e-16
```

Interpretation

Girth and height are associated with an increase in tree volume.

The regression coefficients show the expected change in tree volume associated with each predictor while holding the other predictor constant.

Example 2: Penguins

We will use the `penguins` dataset from the `palmerpenguins` package.

Data Visualisation

```
library(palmerpenguins)
```

```
##
## Attache Paket: 'palmerpenguins'

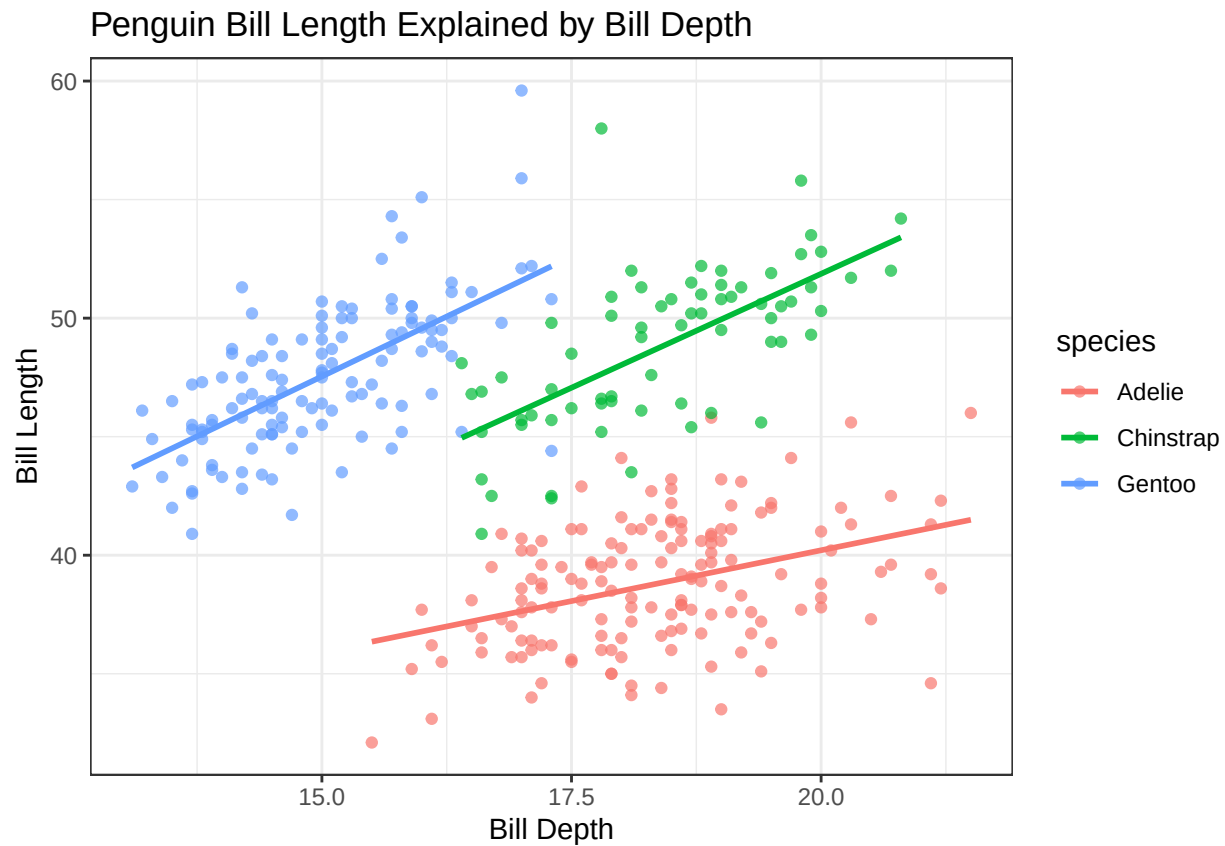
## Die folgenden Objekte sind maskiert von 'package:datasets':
##
##      penguins, penguins_raw
```

```
penguins |>
  ggplot(aes(bill_depth_mm, bill_length_mm)) +
  geom_point(aes(color = species), alpha = 0.7) +
  geom_smooth(
    aes(color = species),
    method = "lm",
    se = FALSE
  ) +
  theme_bw() +
  labs(
    title = "Penguin Bill Length Explained by Bill Depth",
    x = "Bill Depth",
    y = "Bill Length"
  )
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 2 rows containing non-finite outside the scale range
## ('stat_smooth()').
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## ('geom_point()').
```



Model

We can use multiple linear regression to model bill length using bill depth and species as predictors.

```
model2 <- lm(
  bill_length_mm ~ bill_depth_mm + species,
  data = penguins
)

summary(model2)
```

```
##
## Call:
## lm(formula = bill_length_mm ~ bill_depth_mm + species, data = penguins)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -8.0300 -1.5828  0.0733  1.6925 10.0313
```

```
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    13.2164     2.2475   5.88 9.83e-09 ***
## bill_depth_mm     1.3940     0.1220  11.43 < 2e-16 ***
## speciesChinstrap  9.9390     0.3678  27.02 < 2e-16 ***
## speciesGentoo    13.4033     0.5118  26.19 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.518 on 338 degrees of freedom
## (2 Beobachtungen als fehlend gelöscht)
## Multiple R-squared:  0.7892, Adjusted R-squared:  0.7874
## F-statistic: 421.9 on 3 and 338 DF, p-value: < 2.2e-16
```

Interpretation

Approximately **79% of the variation in bill length** can be explained by bill depth and species, based on the R-squared value.

Using **Adelie** as the reference species:

- Changing from Adelie to Chinstrap is associated with an estimated increase of **9.939 mm** in bill length, holding bill depth constant.
- Changing from Adelie to Gentoo is associated with an estimated increase of **13.4 mm** in bill length, holding bill depth constant.

Summary

Multiple linear regression allows us to examine the relationship between an outcome variable and several predictors.

In these examples:

- **Trees:** Volume is the outcome, while Girth and Height are the predictors.
- **Penguins:** Bill Length is the outcome, while Bill Depth and Species are the predictors.